

MIP_PROMPT_LIBRARY_V2_PART6_MODERNIZATION_INTEL

Mainframe Intelligence Platform

Enterprise Prompt Library V2

Part 6 - Modernization Intelligence & Transformation Planning

Version: 2.0

Status: Approved

Purpose

Most modernization initiatives fail because organizations begin with technology decisions.

Examples:

- Convert COBOL to Java
- Move to cloud
- Build microservices
- Rewrite everything

These decisions are often made before the system is understood.

MIP reverses this process.

Repository

↓

Discovery

↓

Metadata

↓

Knowledge Graph

↓

Validation

↓

Reasoning

↓

Modernization Intelligence

↓

Transformation Roadmap

Modernization becomes evidence-driven rather than opinion-driven.

Modernization Principles

Understand Before Transform

Never modernize what is not understood.

Preserve Business Value

Business rules are assets.

Technology is implementation.

Reduce Risk Incrementally

Prefer evolutionary modernization over big-bang rewrites.

Optimize for Business Outcomes

Technology changes must support measurable business goals.

Explain Every Recommendation

Every recommendation must be supported by metadata and graph evidence.

Prompt 31

Business Capability Discovery

File

prompts/modernization/business_capability_discovery.md

Purpose

Identify business capabilities hidden within legacy systems.

Context

Programs rarely map directly to business functions.

Capabilities emerge from groups of programs, jobs, datasets, and tables.

Inputs

- Program Metadata
- Job Metadata
- Data Lineage
- Knowledge Graph

Instructions

Identify:

- Business Functions
- Business Processes
- Shared Capabilities
- Operational Domains

Group related technical assets into capability clusters.

Expected Output

Capability Catalog

| Capability | Description | Assets |

Capability Map

Criticality Ranking

Constraints

Use evidence-based clustering.

Success Criteria

Capabilities become visible and traceable.

Prompt 32

Domain Boundary Discovery

File

prompts/modernization/domain_boundary_discovery.md

Purpose

Identify candidate business domains.

Context

Many legacy applications evolved into monoliths.

Domain boundaries are hidden.

Inputs

Knowledge Graph

Instructions

Analyze:

- Program clusters
- Data ownership
- Dependency density
- Execution flows

Identify potential domains.

Expected Output

Domain Catalog

Domain Ownership Matrix

Cross-Domain Dependencies

Success Criteria

Logical business domains emerge.

Prompt 33

Service Candidate Discovery

File

prompts/modernization/service_candidate_discovery.md

Purpose

Identify microservice candidates.

Context

Services should emerge from business capabilities.

Not from arbitrary code decomposition.

Inputs

Domains

Capabilities

Knowledge Graph

Instructions

Identify candidate services based on:

- Cohesion
- Ownership
- Dependency Boundaries
- Data Boundaries

Expected Output

Service Candidate Catalog

Dependency Analysis

Migration Complexity Score

Success Criteria

Service candidates are evidence-based.

Prompt 34

API Candidate Discovery

File

prompts/modernization/api_candidate_discovery.md

Purpose

Identify APIs that can expose legacy functionality.

Context

API enablement is often the fastest modernization path.

Inputs

Program Metadata

Transaction Metadata

Business Capabilities

Instructions

Identify:

- Reusable Functions
- Shared Operations
- Common Transactions

Generate API candidates.

Expected Output

API Catalog

Input/Output Contracts

Consumer Analysis

Success Criteria

High-value APIs become visible.

Prompt 35

Event Candidate Discovery

File

prompts/modernization/event_candidate_discovery.md

Purpose

Identify event-driven modernization opportunities.

Context

Many legacy systems contain natural business events.

Inputs

Programs

Jobs

Tables

Lineage

Instructions

Identify events such as:

- Account Opened
- Card Activated
- Payment Processed
- Claim Approved

Map event producers and consumers.

Expected Output

Event Catalog

Producer/Consumer Matrix

Event Flow Diagram

Success Criteria

Event-driven architecture opportunities are identified.

Prompt 36

Strangler Pattern Planning

File

prompts/modernization/strangler_planning.md

Purpose

Design incremental modernization strategies.

Context

Big-bang rewrites carry significant risk.

Inputs

Capabilities

Domains

Service Candidates

Instructions

Identify:

- Isolation Points
- Façade Opportunities
- Incremental Migration Paths

Generate strangler strategy.

Expected Output

Strangler Roadmap

Migration Phases

Risk Analysis

Success Criteria

Incremental modernization path exists.

Prompt 37

Technical Debt Analysis

File

prompts/modernization/technical_debt_analysis.md

Purpose

Quantify modernization pain points.

Context

Not all legacy code is problematic.

Focus modernization effort where value exists.

Inputs

Metadata

Graph

Complexity Metrics

Instructions

Identify:

- Highly Coupled Components
- Dead Code
- Circular Dependencies
- Obsolete Technologies

Expected Output

Technical Debt Report

Debt Ranking

Remediation Priorities

Success Criteria

Debt becomes measurable.

Prompt 38

Migration Wave Planning

File

prompts/modernization/migration_wave_planning.md

Purpose

Create phased modernization roadmap.

Context

Large enterprises modernize over multiple years.

Inputs

Capabilities

Domains

Risk Analysis

Instructions

Generate:

- Wave 1
- Wave 2
- Wave 3
- Wave 4

Prioritize:

- Business Value
- Complexity
- Risk

Expected Output

Migration Roadmap

Wave Plans

Dependencies

Success Criteria

Transformation becomes executable.

Prompt 39

Risk-Aware Modernization Assessment

File

prompts/modernization/risk_aware_assessment.md

Purpose

Evaluate modernization risk.

Context

Risk is often underestimated.

Inputs

Knowledge Graph

Lineage

Impact Analysis

Instructions

Assess:

- Operational Risk
- Data Risk
- Regulatory Risk
- Business Continuity Risk

Expected Output

Risk Matrix

Risk Rankings

Mitigation Recommendations

Success Criteria

Leadership understands modernization risk.

Prompt 40

Enterprise Transformation Blueprint

File

prompts/modernization/transformation_blueprint.md

Purpose

Generate a complete modernization blueprint.

Context

Leadership requires an actionable plan, not just technical analysis.

Inputs

All MIP Outputs

Instructions

Produce:

- Current State Architecture
- Future State Architecture
- Capability Map
- Domain Map
- Service Landscape
- API Strategy
- Event Strategy
- Migration Roadmap
- Risk Assessment
- Investment Priorities

Expected Output

Executive Blueprint

Architecture Views

Transformation Roadmap

Investment Guidance

Success Criteria

Leadership can use the output to drive modernization decisions.

Enterprise Modernization Outcomes

After completing Part 6, MIP should answer:

- What business capabilities exist?
 - What domains exist?
 - What services should be created?
 - Which APIs should be exposed?
 - Which events should be emitted?
 - What should be modernized first?
 - What is the safest migration path?
 - What is the future-state architecture?
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Prompt Library Completion Criteria

At the completion of Parts 1–6, MIP should be capable of:

- ✓ Repository Discovery
 - ✓ Metadata Extraction
 - ✓ Knowledge Graph Construction
 - ✓ Validation & Quality Assurance
 - ✓ Reasoning & Impact Analysis
 - ✓ Modernization Intelligence
 - ✓ Transformation Planning
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MIP Differentiator

Most tools answer:

"What code exists?"

MIP answers:

"What should the organization do next?"

End of Part 6.